WORKSHEET: MODEL EVALUATION TECHNIQUES

Topic: Machine Learning / Metrics | Duration: 60 Minutes

Name: _____ Roll Number: _____ Date: _____

Why Evaluate?

Building a model is like studying for an exam. **Evaluation** is taking the exam to see how well you actually learned.

Different problems need different "Exams" (Metrics).

Part 1: The Accuracy Paradox (Why simple accuracy fails)

Scenario: We built a "Lazy" model to predict Diabetes. In our town, most people are healthy.

The Model's Logic: It simply predicts **"Healthy (0)"** for everyone. It doesn't even look at the data!

ID	Weight (kg)	Glucose	Actual Diagnosis	Model Prediction	Check
#1	70	90	Healthy (0)	Healthy (0)	Correct
#2	65	85	Healthy (0)	Healthy (0)	Correct
#3	80	100	Healthy (0)	Healthy (0)	Correct
#4	75	95	Healthy (0)	Healthy (0)	Correct
#5	68	88	Healthy (0)	Healthy (0)	Correct
#6	72	92	Healthy (0)	Healthy (0)	Correct
#7	78	98	Healthy (0)	Healthy (0)	Correct
#8	66	86	Healthy (0)	Healthy (0)	Correct
#9	74	94	Healthy (0)	**Diabetic (1)**	**WRONG (FP)**
#10	**105**	**220**	**Diabetic (1)**	**Healthy (0)**	**WRONG (FN)**

Step 1: Calculate "Accuracy"

Total Patients = 10. Correct Guesses = 8.

$$Accuracy = \frac{8}{10} = \mathbf{80\%}$$

Analysis: 80% looks decent (B grade). But is this model useful?

NO! It killed the sick patient (FN) and scared a healthy one (FP).

This is why we need the **Confusion Matrix**—to separate the "Easy Corrects" from "Critical Failures".

	Predicted: Diabetic (1)	**Predicted: Healthy (0)**
Actual: Diabetic (1)	**True Positive (TP) = 0** (Caught None)	**False Negative (FN) = 1** (Patient #10 - Missed!)
Actual: Healthy (0)	**False Positive (FP) = 1** (Patient #9 - False Alarm)	**True Negative (TN) = 8**

Step 3: The Deep Analysis (Breakdown by Category)

Accuracy is just an average. Let's grade the model for each category separately:

Category 1: The Sick (Real Positives)

- Total Sick: 1
- **TPR (True Positive Rate):** Accuracy for Sick people.
  $\frac{TP}{TotalSick} = \frac{0}{1} = \mathbf{0\%}$
  *(We call this **Recall**)*
- **FNR (False Negative Rate):** The Miss Rate.
  $\frac{FN}{TotalSick} = \frac{1}{1} = \mathbf{100\%}$

Category 0: The Healthy (Real Negatives)

- Total Healthy: 9
- **TNR (True Negative Rate):** Accuracy for Healthy people.
  $\frac{TN}{TotalHealthy} = \frac{8}{9} \approx \mathbf{89\%}$
  *(We call this **Specificity**)*
- **FPR (False Positive Rate):** False Alarm Rate.
  $\frac{FP}{TotalHealthy} = \frac{1}{9} \approx \mathbf{11\%}$

The Paradox Revealed:

Total Accuracy (80%) is just an average of 89% (Healthy) and 0% (Sick).

Step 4: Standard Metrics Simplified

- **Recall (TPR):** 0%. (Critical for Medical/Safety). We missed the patient.
- **Precision:** $\frac{TP}{TP+FP} = \frac{0}{0+1} = \mathbf{0\%}$. (Of the people we *said* were sick, how many actually were? 0.)

Quick Check 1: In a dataset with 99 healthy people and 1 sick person, if a model predicts everyone as "Healthy", what is its accuracy?

- 1%
- 50%
- 99%
- 100%

Part 2: Regression Metrics (Predicting Numbers)

Used when the answer is a number (e.g., Price of a House).

> ***The Analogy: "The Darts Game"***
> *You want your darts (predictions) to be close to the bullseye (actual value).*

Imagine we are predicting the prices of 5 houses (in Thousands of Dollars):

House	Actual Price (y)	Predicted Price ($\hat{y}$)	Error ($y - \hat{y}$)	Absolute Error ($\lvert y - \hat{y} \rvert$)	Squared Error ($(y - \hat{y})^2$)
#1	100	90	10	10	100
#2	150	170	-20	20	400
#3	200	180	20	20	400
#4	250	240	10	10	100
#5 (Outlier)	500	400	100	100	10,000
TOTAL	**-**	**-**	**-**	**160**	**11,000**

Step-by-Step Calculations:

1. **MAE (Mean Absolute Error):** Average of absolute errors.

$$MAE = \frac{10 + 20 + 20 + 10 + 100}{5} = \frac{160}{5} = \mathbf{32}$$

2. **MSE (Mean Squared Error):** Average of squared errors.

$$MSE = \frac{100 + 400 + 400 + 100 + 10000}{5} = \frac{11000}{5} = \mathbf{2200}$$

3. **RMSE (Root Mean Squared Error):** Standard deviation of residuals.

$$RMSE = \sqrt{2200} \approx \mathbf{46.9}$$

4. R^2 **(R-Squared):** The "Fitness" Score.

> **1. TSS (Total Sum of Squares):** The "Total Error" if we just guessed the average ($\bar{y}$).
>
> $$TSS = \sum (y - \bar{y})^2$$
>
> In our case (Mean = 240): $(100 - 240)^2 + \ldots + (500 - 240)^2 = \mathbf{97,000}$

> **2. RSS (Residual Sum of Squares):** The "Remaining Error" after using our model ($\hat{y}$).

$$RSS = \sum (y - \hat{y})^2$$

In our case: $100 + 400 + 400 + 100 + 10000 = \mathbf{11,000}$

$$R^2 = 1 - \frac{RSS}{TSS} = 1 - \frac{11,000}{97,000} \approx \mathbf{0.88}$$

(Interpretation: Our model explained 88% of the total variation!)

Insight 1: MAE vs MSE

House #5 (Outlier) error was 100.

- **MAE:** Contributed 100 to sum.
- **MSE:** Contributed 10,000 to sum!

"MAE ignores outliers, MSE punishes large errors."

Insight 2: Why use RMSE?

MSE gives results in "Dollars Squared" (^{2}), which humans can't visualize.

"RMSE brings the 'penalty' of MSE back to real units ($ Dollars)."

In our example, an RMSE of 46.9 means our predictions are typically off by ~$47k.

Insight 3: The R^2 Logic

If you knew nothing about ML, you'd just guess the **Average Price ($240k)** for every house.

"R^2 tells you what % of errors you 'fixed' compared to the Average baseline."

Our 0.88 means we reduced the variance by 88%.

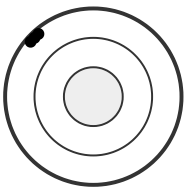
Metric	Concept	Formula
MAE	Average "Distance" from truth. Simple to understand ($stands for \pm 32$ units).	$\frac{1}{n}\sum \lvert y - \hat{y} \rvert$
MSE	Punishes large errors heavily (Squaring). Harder to explain units (2200 $units^2$?).	$\frac{1}{n}\sum (y - \hat{y})^2$
RMSE	Punishes large errors, but brings units back to normal (≈ 46.9 units).	$\sqrt{MSE}$
R^2	"Score" from 0 to 1. How much better is your model than just guessing the average? (1.0 is perfect).	$1 - \frac{RSS}{TSS}$

Quick Check 2: Fill in the blanks.

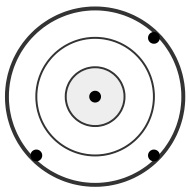
MAE is better if we want to ignore _____, while MSE is better if we want to punish _____ errors.

Part 3: Bias vs Variance (The Archer)

This explains *why* a model makes mistakes.



High Bias (Underfitting)
Consistent, but consistently Wrong.



High Variance (Overfitting)
Calculated average is right, but shots are everywhere.

Problem	The "Model" Reason	The "Data" Reason
High Bias (Underfitting)	**Too Simple:** The model isn't "smart" enough to find the pattern (e.g., using a linear model for complex data).	**Missing Info:** Not enough useful features were provided to help the model learn.
High Variance (Overfitting)	**Too Complex:** The model is "too flexible" and starts memorizing noise/ $unimportant$ details.	**Too Little Data:** Small datasets make the model think unique "flukes" are actually rules.

The Tradeoff:

- **Complex Model:** Low Bias, High Variance (Memorizes noise).
- **Simple Model:** High Bias, Low Variance (Oversimplifies reality).
- **Goal:** Find the "Sweet Spot" in the middle.

Quick Check 3: A model that performs 100% on training data but only 60% on test data is likely suffering from:

- High Bias (Underfitting)
- High Variance (Overfitting)
- Low Bias, Low Variance

Part 4: Model Validation (Cross-Validation)

The Limit of "Train/Test Split": If you just split data once, you might get lucky (or unlucky).

The Analogy: "The Rotation Policy"
Instead of testing on the same chapter every time, rotating the chapters ensures you know the whole book.

K-Fold Cross Validation (e.g., K=5):

Test	Train

Train	Test	Train

Train	Test	Train

... Repeat K times, then Average the Scores.

Benefit: It gives a much more reliable estimate of how your model will perform on new data because every data point gets a chance to be in the "Test Set".

Final Challenge: Cross-Validation

In 10-fold cross-validation, the model is trained and tested _____ times.